Part 1: Formulation

1 Decision Variables

Define decision variable:

- X_{ijklm}: Quantity (tons) of commodity i purchased from country j, transported via port k, stored in warehouse l, delivered to camp m.

2 Objective Function

Minimize total cost:

$$\min \sum_{ijklm} X_{ijklm} \times (C_{procurement} + C_{transport_{jk}} + C_{handling_k} + C_{transport_{kl}} + C_{handling_l} + C_{transport_{lm}}) \tag{1}$$

where:

- $C_{procurement}$: Procurement cost per ton from j.
- $C_{transport_{jk}}$: Transportation cost per ton from j to port k.
- $C_{handling_k}$: Handling cost per ton at port k.
- $C_{transport_{kl}}$: Transportation cost per ton from port k to warehouse l.
- $C_{handling_l}$: Handling cost per ton at warehouse l.
- $C_{transport_{lm}}$: Transportation cost per ton from warehouse l to camp m.

3 Constraints

3.1 Procurement Constraints

$$\sum_{iklm} X_{ijklm} \leq P_j, \quad \forall j \tag{2}$$

3.2 Port Capacity Constraints

$$\sum_{ijlm} X_{ijklm} \leq T_k, \quad \forall k \tag{3}$$

3.3 Warehouse Capacity Constraints

$$\sum_{ijkm} X_{ijklm} \leq W_l, \quad \forall l \tag{4}$$

3.4 Nutritional Requirements

$$\sum_{ijkt} X_{ijklm} \times N_i \geq D_m, \quad \forall m \tag{5}$$

3.5 Non-Negativity Constraint

$$X_{ijklm} \geq 0, \quad \forall i, j, k, l, m \tag{6}$$

4 Summary

This formulation optimizes procurement and transportation costs while ensuring demand constraints are met. It provides an optimal allocation of resources for efficient food distribution to beneficiary camps.